

EDUCATION

McGill University

Sept. 2023 - present

PhD Candidate in Mathematics and Statistics

- Supervisor: Dr. Elliot Paquette
- Research Topics: random matrix theory, stochastic processes, high-dimensional probability and optimization

McGill University

2020-2023

Master of Science in Mathematics and Statistics

- Supervisor: Dr. Adam Oberman
- Cumulative GPA: 4.0/4.0
- Associated with Mila – Quebec Artificial Intelligence Institute

The University of British Columbia

2015-2019

Bachelor of Science - Honours in Mathematics and Minor in Data Science

- Cumulative GPA: 4.33/4.33

RESEARCH EXPERIENCE

Microsoft + Nuance

2023 Summer

NLP Research Intern

- Implemented generative AI tools for customer service
- Augmented large language models using a knowledge base retriever

Phyla

2021 Summer

Machine Learning Research Intern

- Developed a graph neural network for classification of irritable bowel diseases (IBD)
- Implemented a pipeline to run, test, and log experimental results
- Developed a custom cross validation method to reduce the impact of batch effects

The University of British Columbia

2020-2021

Research Assistant 2

- Analyzed and developed a differential equation model coupling the dynamics of opinion and disease
- Find that more reactive populations can experience smaller peak epidemic size, but more epidemic waves and more cases

Mathematics and Climate Research Network

2019-2021

Student Researcher

- Collaborated with an international group of students and professors to develop a novel data assimilation method
- Investigated the use of model and data dimension reduction methods to improve the effectiveness of particle filters

PUBLICATIONS

- Elliot Paquette, **Noah Marshall**, Lucas Benigni, Guangyuan Wang, Atish Agarwala, and Courtney Paquette. 2026. Phases of Muon: When Muon Eclipses SignSGD. *Preprint* (arXiv:2605.09552)
- Ke Liang Xiao, **Noah Marshall**, Atish Agarwala, and Elliot Paquette. 2025. Exact Risk Curves of signSGD in High-Dimensions: Quantifying Preconditioning and Noise-Compression Effects. *International Conference on Machine Learning*
- **Noah Marshall**, Ke Liang Xiao, Atish Agarwala, and Elliot Paquette. 2025. To Clip or Not to Clip: the Dynamics of SGD with Gradient Clipping in High-Dimensions. *International Conference on Learning Representations*
- Tiago Salvador, Stephanie Cairns, Vikram Voleti, **Noah Marshall**, and Adam Oberman. 2022. FairCal: Fairness Calibration for Face Verification. *International Conference on Learning Representations*
- Rebecca C. Tyson, **Noah D. Marshall**, and Bert O. Baumgaertner. 2022. Transient prophylaxis and multiple epidemic waves. *AIMS Mathematics*
- Aishah Albarakati, Marko Budišić, Rose Crocker, Juniper Glass-Klaiber, Sarah Iams, John Maclean, **Noah Marshall**, Colin Roberts, and Erik S. Van Vleck.* 2022. Model and Data Reduction for Data Assimilation: Particle Filters Employing Projected Forecasts and Data with Application to a Shallow Water Model. *Computers & Mathematics with Applications* (*equal contribution, sorted by last name)

SKILLS

MATHEMATICS	Probability stochastic processes random matrix theory high-dimensional statistics optimization
MODELLING	Covariance estimation in high dimensions particle filters & data assimilation optimizer & training dynamics scaling laws

TEACHING AND MENTORSHIP

Course Instructor 2026

McGill University

- Instructor for Intermediate Calculus

Directed Reading Program 2021-present

McGill University

- Lead a mentored reading program for undergraduate students
- Topics: Gaussian processes on manifolds, generative diffusion models, probabilistic graphical models, tree indexed stochastic processes, percolations on neural networks

Graduate Teaching Assistant 2022-present

McGill University

- Introduction to stochastic processes, Calculus 1

AWARDS

Canadian Graduate Scholarship (NSERC) present

- Program provides financial support to high-calibre students

Institut des sciences mathématiques Graduate Scholarship present

- Awarded to outstanding PhD students in mathematics

NSERC Undergraduate Student Research Award 2018

- Funding awarded to undergraduate students pursuing summer research in natural sciences or engineering